

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: **Cyber & Information Security**
Practical: 3 Message Authentication Codes

AIM: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity

Code :

```
import hmac
import hashlib

# User input
secret_key = input("Enter the secret key: ").encode()
message = input("Enter the message: ").encode()

# Generate MAC
mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:", mac)

# Verification
received_message = input("\nEnter the message for verification: ").encode()

# Generate MAC for the received message
received_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

print("\nVerifying Message...")

if hmac.compare_digest(mac, received_mac):
    print("Authentication Successful!")
    print("Message Integrity Verified.")
else:
    print("Authentication Failed!")
    print("Message has been modified.")
```

NAME :NAME: ADARSH SHUKLA

ROLL NO :NO: T117

Sub :Sub: **Cyber & Information Security**

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: **Cyber & Information Security**
Practical: 3 Message Authentication Codes

```
*a.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/a.py (3.14.6)*
File Edit Format Run Options Window Help

import hmac
import hashlib

secret_key = input("Enter the secret key: ").encode()
message = input("Enter the message: ").encode()

mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC:", mac)

received_message = input("\nEnter the message for verification: ").encode()

received_mac = hmac.new(secret_key, received_message, hashlib.sha256).hexdigest()

print("\nVerifying Message...")

if hmac.compare_digest(mac, received_mac):
    print("Authentication Successful!")
    print("Message Integrity Verified.")
else:
    print("Authentication Failed!")
    print("Message has been modified.")

Ln: 4 Col: 0
```

Output :

NAME :NAME: ADARSH SHUKLA
ROLL NO :NO: T117
Sub :Sub: **Cyber & Information Security**

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical: 3 Message Authentication Codes



```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-49/AppData/Local/Programs/Python/Python314/a.py =====
Enter the secret key: T117
Enter the message: adarsh

Generated MAC: b5dc8d105cbaba68ef43ffc45614314c31068b39d26a8a213ec37083ef5c8d8c

Enter the message for verification: adarsh

Verifying Message...
Authentication Successful!
Message Integrity Verified.

>>>
===== RESTART: C:/Users/IT-49/AppData/Local/Programs/Python/Python314/a.py =====
Enter the secret key: T117
Enter the message: adarsh shukla

Generated MAC: a97149d9d295cd4149c83f67fa59702693118236a1a4bbd590foe988ac8c243d

Enter the message for verification: adarsh

Verifying Message...
Authentication Failed!
Message has been modified.

>>>
```

Gui :

Code :

```
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Function to generate MAC
def generate_mac():
    key = entry_key.get().encode()
    message = entry_message.get().encode()

    if not key or not message:
        messagebox.showerror("Error", "Please enter both key and message")
```

NAME :NAME: ADARSH SHUKLA

ROLL NO :NO: T117

Sub :Sub: Cyber & Information Security

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical: 3 Message Authentication Codes

```
    return

    mac = hmac.new(key, message, hashlib.sha256).hexdigest()
    entry_mac.delete(0, tk.END)
    entry_mac.insert(0, mac)

# Function to verify MAC
def verify_mac():
    key = entry_key.get().encode()
    message = entry_verify_message.get().encode()
    mac_original = entry_mac.get()

    if not key or not message or not mac_original:
        messagebox.showerror("Error", "Please fill all fields")
        return

    new_mac = hmac.new(key, message, hashlib.sha256).hexdigest()

    if hmac.compare_digest(mac_original, new_mac):
        messagebox.showinfo("Result", "✅ Authentication Successful!\nMessage Integrity Verified.")
    else:
        messagebox.showerror("Result", "❌ Authentication Failed!\nMessage has been modified.")

# GUI Window
root = tk.Tk()
root.title("MAC Generator & Verifier")
root.geometry("500x400")

# Labels & Entries
tk.Label(root, text="Secret Key").pack()
entry_key = tk.Entry(root, width=50)
entry_key.pack()

tk.Label(root, text="Message (Generate MAC)").pack()
entry_message = tk.Entry(root, width=50)
entry_message.pack()

tk.Button(root, text="Generate MAC", command=generate_mac, bg="lightblue").pack(pady=5)

tk.Label(root, text="Generated MAC").pack()
```

NAME :NAME: ADARSH SHUKLA

ROLL NO :NO: T117

Sub :Sub: Cyber & Information Security

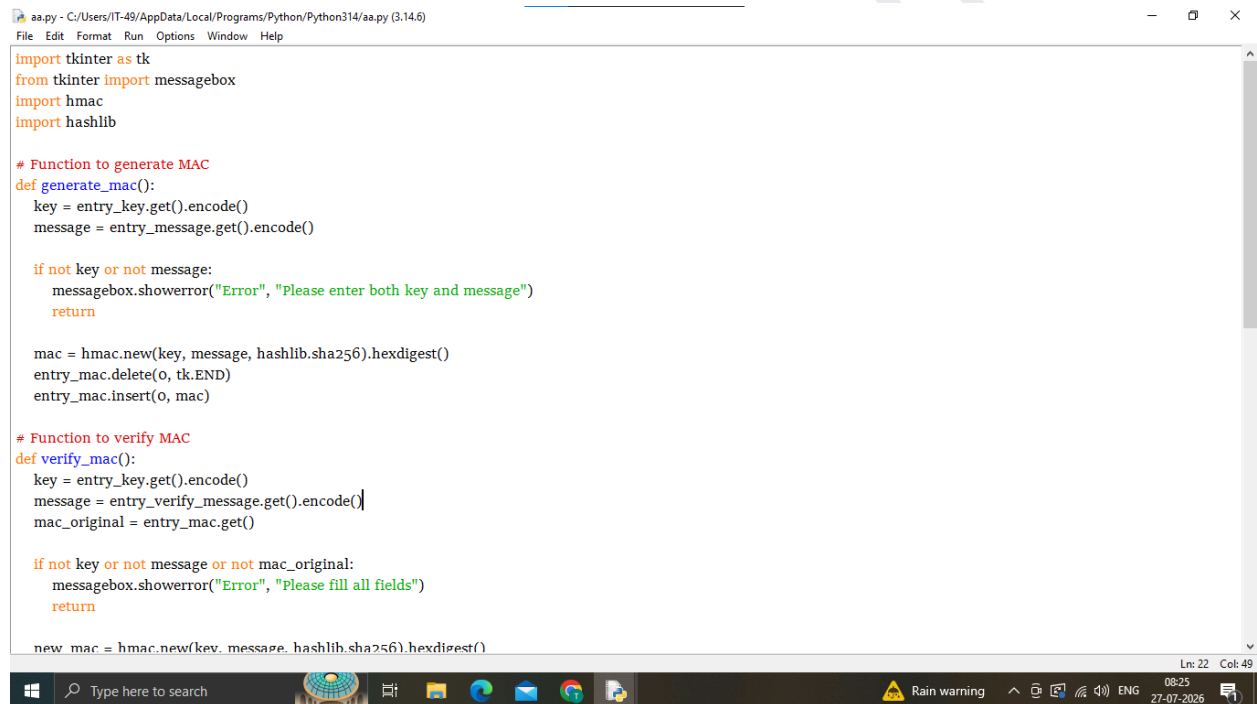
SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical: 3 Message Authentication Codes

```
entry_mac = tk.Entry(root, width=50)
entry_mac.pack()
```

```
tk.Label(root, text="Message (Verify)").pack()
entry_verify_message = tk.Entry(root, width=50)
entry_verify_message.pack()
```

```
tk.Button(root, text="Verify MAC", command=verify_mac, bg="lightgreen").pack(pady=10)
```

```
# Run GUI
root.mainloop()
```



```
aa.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/aa.py (3.14.6)
File Edit Format Run Options Window Help

import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

# Function to generate MAC
def generate_mac():
    key = entry_key.get().encode()
    message = entry_message.get().encode()

    if not key or not message:
        messagebox.showerror("Error", "Please enter both key and message")
        return

    mac = hmac.new(key, message, hashlib.sha256).hexdigest()
    entry_mac.delete(0, tk.END)
    entry_mac.insert(0, mac)

# Function to verify MAC
def verify_mac():
    key = entry_key.get().encode()
    message = entry_verify_message.get().encode()
    mac_original = entry_mac.get()

    if not key or not message or not mac_original:
        messagebox.showerror("Error", "Please fill all fields")
        return

    new_mac = hmac.new(key, message, hashlib.sha256).hexdigest()
```

NAME :NAME: ADARSH SHUKLA
ROLL NO :NO: T117
Sub :Sub: Cyber & Information Security

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: **Cyber & Information Security**
Practical: 3 Message Authentication Codes

```
aa.py - C:/Users/IT-49/AppData/Local/Programs/Python/Python314/aa.py (3.14.6)
File Edit Format Run Options Window Help

if hmac.compare_digest(mac_original, new_mac):
    messagebox.showinfo("Result", "✅ Authentication Successful!\nMessage Integrity Verified.")
else:
    messagebox.showerror("Result", "❌ Authentication Failed!\nMessage has been modified.")

# GUI Window
root = tk.Tk()
root.title("MAC Generator & Verifier")
root.geometry("500x400")

# Labels & Entries
tk.Label(root, text="Secret Key").pack()
entry_key = tk.Entry(root, width=50)
entry_key.pack()

tk.Label(root, text="Message (Generate MAC)").pack()
entry_message = tk.Entry(root, width=50)
entry_message.pack()

tk.Button(root, text="Generate MAC", command=generate_mac, bg="lightblue").pack(pady=5)

tk.Label(root, text="Generated MAC").pack()
entry_mac = tk.Entry(root, width=50)
entry_mac.pack()

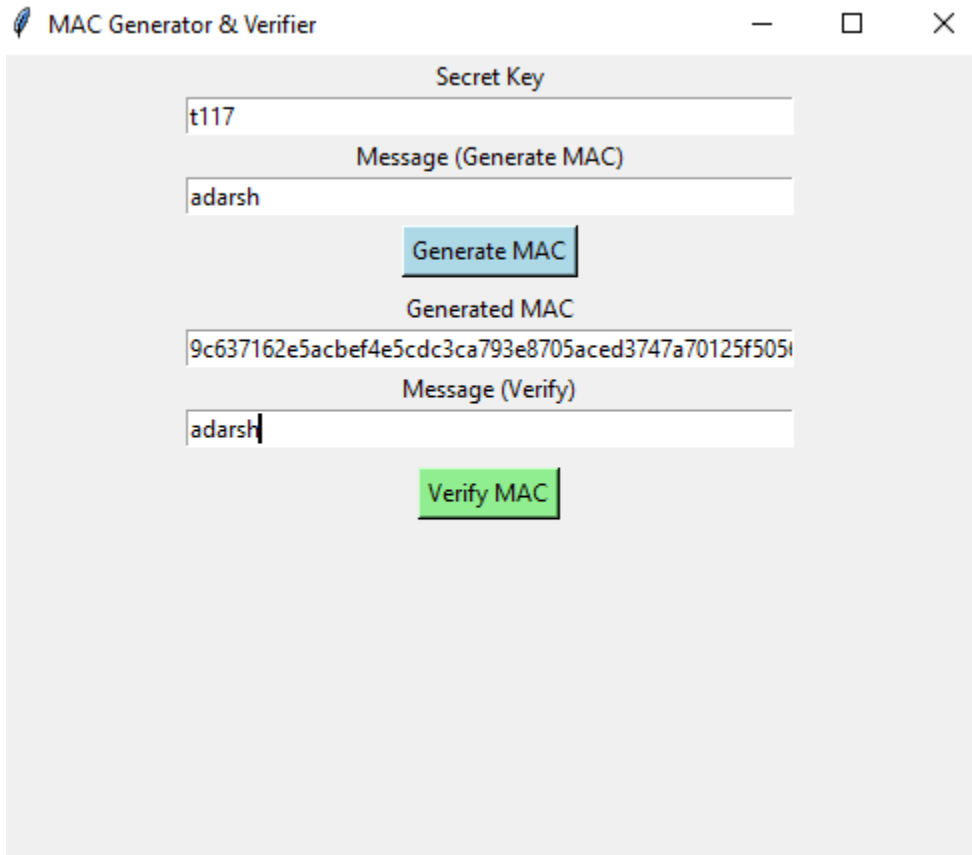
tk.Label(root, text="Message (Verify)").pack()
entry_verify_message = tk.Entry(root, width=50)
entry_verify_message.pack()

Ln: 22 Col: 49
```

Output :

NAME :NAME: ADARSH SHUKLA
ROLL NO :NO: T117
Sub :Sub: **Cyber & Information Security**

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical: 3 Message Authentication Codes



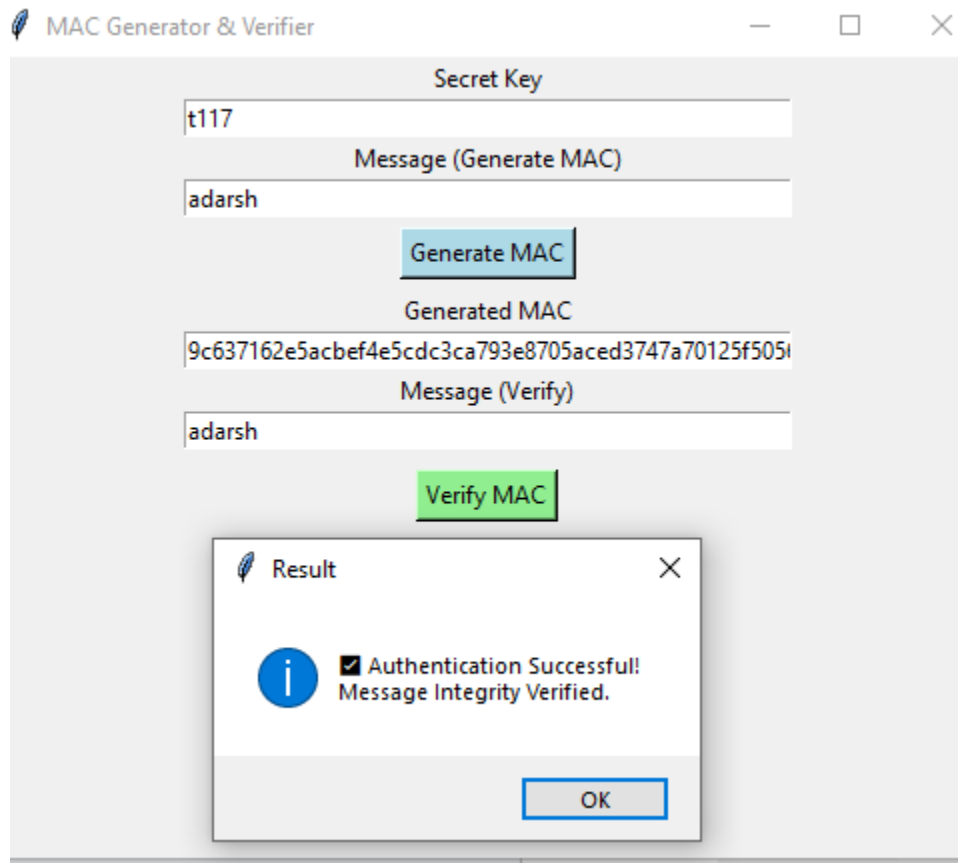
The screenshot shows a window titled "MAC Generator & Verifier". It contains the following fields and buttons:

- Secret Key:** A text input field containing "t117".
- Message (Generate MAC):** A text input field containing "adarsh".
- Generate MAC:** A blue button.
- Generated MAC:** A text input field containing the hexadecimal string "9c637162e5acbef4e5cdc3ca793e8705aced3747a70125f505d".
- Message (Verify):** A text input field containing "adarsh".
- Verify MAC:** A green button.

A large, diagonal watermark "T117 adarsh" is visible across the lower half of the image.

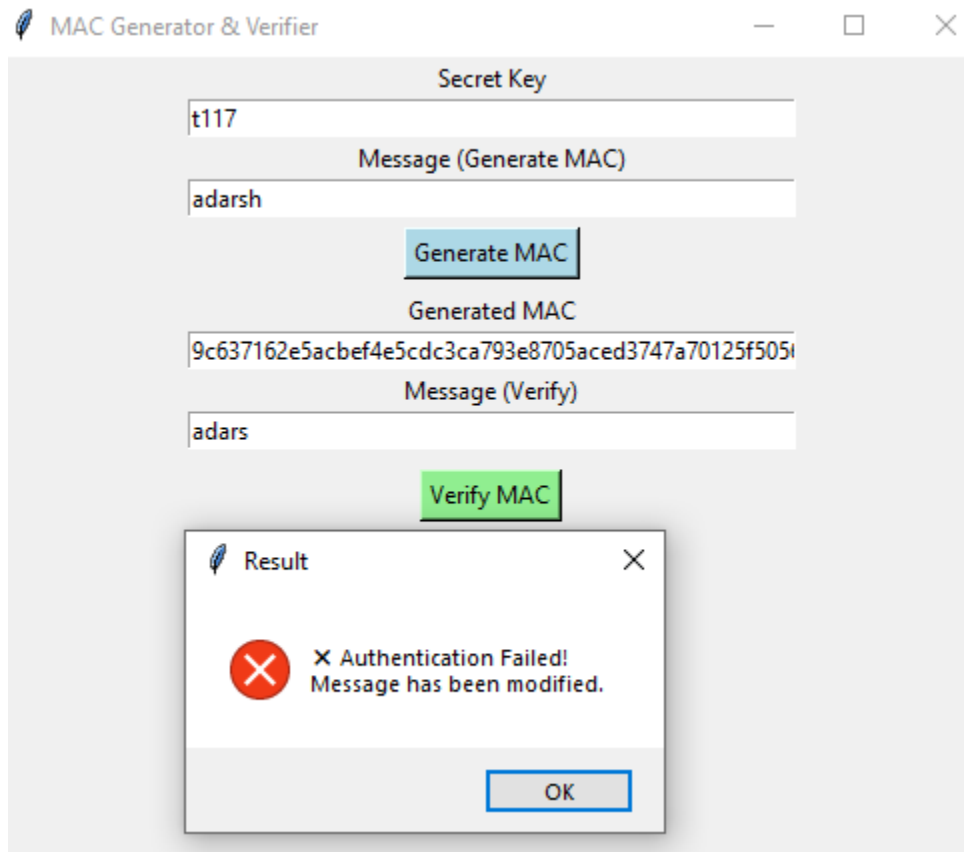
NAME :NAME: ADARSH SHUKLA
ROLL NO :NO: T117
Sub :Sub: Cyber & Information Security

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical: 3 Message Authentication Codes



NAME :NAME: ADARSH SHUKLA
ROLL NO :NO: T117
Sub :Sub: Cyber & Information Security

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Cyber & Information Security
Practical: 3 Message Authentication Codes



NAME :NAME: ADARSH SHUKLA
ROLL NO :NO: T117
Sub :Sub: Cyber & Information Security